



INL 261

PART E OF E

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Table of Contents

INTRODUCTION	2
PROJECT JOURNEY	2
CHALLENGES FACED	4
LESSONS LEARNED	4
CONCLUSION	5

INTRODUCTION

Our group was given the fun yet difficult task of designing, 3D printing, assembling, and testing a solar-powered vehicle as part of our INL261 module. In addition to helping us apply our academic knowledge to a real-world, hands-on activity, the project aimed to develop critical leadership, teamwork, and innovation abilities.

Designing a car and printing it seemed like a straightforward endeavour at first. However, as we soon learnt, this initiative put our technological proficiency to the test as well as our capacity for productive group collaboration among both online and in-person students.

PROJECT JOURNEY

Part A: Planning and Research

First, our crew came up with a number of ideas for solar-powered vehicles. We looked into designs that struck a balance between speed, stability, and solar energy efficiency. Every participant offered suggestions, and we began modelling the initial prototypes using CAD programs like Tinkercad and Fusion 360.

Part B: Modelling and Design

Although we made our initial design, it had some shortcomings. Some pieces didn't line up properly for the engine and wheels, while others were too thin to print consistently. We were compelled to redesign our car as a result, which proved to be a valuable educational experience. The car's makeover improved its usability and printability.

Part C: The Process of 3D Printing

Going to the lab and printing the car was the responsibility of a group of the physical students. They initially ran into problems, including structural flaws, scaling problems, and print failures. Rather than give up, they collaborated to troubleshoot. Before the pieces printed perfectly, they verified tolerances, reproduced parts, and changed slicer settings.

Part D: Constructing and Examining

When the components were prepared, the team put the automobile together by fastening the wheels, axles, motor, and solar panel. The automobile performed as expected throughout testing conducted both indoors (with limited success) and outdoors in direct sunlight. For the entire group, it was a satisfying moment to watch the car move for the first time.

At the hangar

List of participants of 3D Printing and assembling:

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Righardt Ferreria 600647 (our Lead Designer, working with fusion 360)

Our team successfully completed a three-week printing process. Throughout we encountered several challenges, particularly regards to the car's size and component placement.

Initially, we faced issues with the switch placement and sizing. To overcome these challenges, we increased the size of the car, switch placement and small gear, made adjustments to ensure a precise fit for every component. We also printed larger wheels to match the revised design. We iterated on our design and printing process, refining our approach with each attempt.

Notably, our printing times improved significantly. The initial print for the chassis, 4 wheels, small gear took 58 minutes 24 seconds. However, after refining our design and settings subsequent prints were much faster with the final wheels and small gear print taking 24 minutes and the entire chassis taking about 48 mins. They were printed separately. We used blue, grey, red, black filament. White being our final colour of the car.

The assembly process was a trial-and-error experience, but through collaboration and creative problem-solving, we developed effective solutions to bring the car to life. The assembling took about 3.5 hours.

I am proud of the progress we made and the teamwork that went into overcoming the challenges we faced.

CHALLENGES FACED

There were both technical and personal difficulties with the project:

Redesigning the Car: We had to start over because our initial design was not print-ready. Although it slowed down our work, this experience taught us how crucial accuracy is when creating 3D objects.

Communication in the Group: This was one of our main problems. It was occasionally challenging to get in touch with everyone because some members worked virtually while others were physically present in the lab. Schedule discrepancies resulted in delays, and occasionally some voices were unheard.

Online vs. Physical Collaboration: There was occasionally conflict due to the division between people who collaborated virtually and those who did not. While online students concentrated more on research, design, and documentation, physical students had direct access to the printer. It took understanding and patience to close this gap.

LESSONS LEARNED

In many respects, the project was a success in spite of the difficulties. In addition to creating a solar automobile, we improved our collaboration abilities.

We discovered that design is an iterative process, and that innovation includes errors and redesigns.

We learnt how crucial it is to communicate clearly when working with both in-person and virtual pupils. Even though on our last days it was still shaky but it was better than in the beginning.

By overcoming printing setbacks instead of giving up on the project, we were able to build resilience.

Above all, we learnt the importance of leadership and teamwork—supporting one another's roles while working towards a common objective.

CONCLUSION

In addition to being a technical challenge, our 3D Printed Solar Car Project tested our ability to work as a team, communicate effectively, and persevere. Even if the automobile we built is modest, the lessons we learnt are significant: creativity necessitates perseverance, teamwork, and leadership.

As a team, we online and in-person students are proud of what we accomplished, and this project has equipped us with both technical knowledge and important life skills.